MATH.APP.220 Multivariable Calculus

Jani Hirvonen

Contents (part 3)

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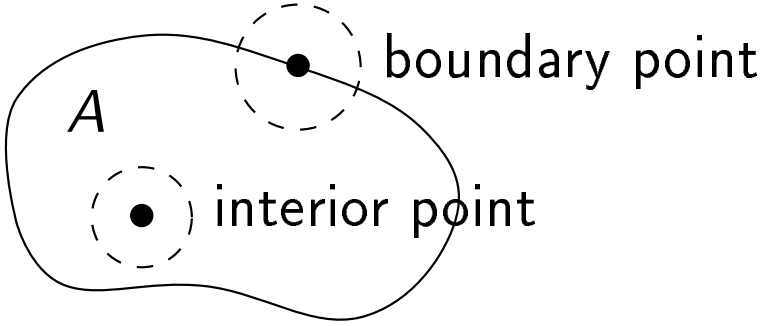
Definition of extrema

- A point $\mathbf{a} \in D$, where $D \subset \mathbb{R}^n$, of the function $f : D \to \mathbb{R}$ is
  - a (global / absolute) **maximum**, if $f(\mathbf{x}) \leq f(\mathbf{a})$ for all $\mathbf{x} \in D$,
  - a (global / absolute) **minimum**, if $f(\mathbf{x}) \geq f(\mathbf{a})$ for all $\mathbf{x} \in D$,
  - a **local maximum**, if $f(\mathbf{x}) \leq f(\mathbf{a})$ in some neighbourhood $\mathbf{x} \in B(\mathbf{a}, r) \cap D$ of $\mathbf{a}$,
  - a **local minimum**, jos $f(\mathbf{x}) \geq f(\mathbf{a})$ in some neighbourhood $\mathbf{x} \in B(\mathbf{a}, r) \cap D$ of $\mathbf{a}$,
- A point $\mathbf{a}$ is an **extremum** (plural: **extrema**), if it is either a maximum or a minimum (plural: maxima, minima) and $f(\mathbf{a})$ is then an **extreme value** either a **maximum value** or a **minimum value**.
- Notations for extreme values: $\max_D f$ and $\min_D f$.

Concepts involving points in a set

- The point $\mathbf{x} \in A$ is an **interior point** of $A \subset \mathbb{R}^n$, if for at least one $r > 0$ it holds that $B(\mathbf{x}, r) \subset A$.
- The point $\mathbf{x}$ is a **boundary point** of $A \subset \mathbb{R}^n$, if for all $r > 0$, the neighbourhood $B(\mathbf{x}, r)$ contains points from both A and its complement A^c, i.e. $B(\mathbf{x}, r) \cap A \neq \emptyset$ and $B(\mathbf{x}, r) \cap A^c \neq \emptyset$ for all $r > 0$. Boundary points can either be in A or A^c.



Parts of a set

- The **boundary** of A is the set of all boundary points of A. It is denoted ∂A. All, some or none of the points in the boundary might be in A.
- The **interior** of A is the set of all interior points of A. It is denoted $\mathrm{int}(A)$.
- The **closure** of A, denoted $\overline{A} = \mathrm{int}(A) \cup \partial A$, contains all interior points and all boundary points (whether the boundary points are in A or not).
- It always holds that $\mathrm{int}(A) \subset A \subset \overline{A}$.

Open and closed sets

- The set A is **closed**, if A contains all of its boundary points.
- The set A is **open**, if A contains none of its boundary points.
- The set A is open, if and only if $A = \mathrm{int}(A)$, and closed, if and only if $A = \overline{A}$.
- For example, $\{(x,y) \in \mathbb{R}^2 : x \geq 0, y \geq 0, x+y \leq 1\}$ is closed, $\{(x,y) \in \mathbb{R}^2 : x > 0, y > 0, x+y < 1\}$ is open, $\{(x,y) \in \mathbb{R}^2 : x \geq 0, y \geq 0, x+y < 1\}$ neither open nor closed.

A bounded set

- The set $A \subset \mathbb{R}^n$ is **bounded**, if $A \subset B(\mathbf{0}, R)$ for (a large number) $R > 0$. Otherwise A is unbounded.
- The **diameter** of a closed and bounded set $A \neq \emptyset$ is

$$\operatorname{diam}(A) = \max\{\|\mathbf{x} - \mathbf{y}\| : \mathbf{x}, \mathbf{y} \in A\}.$$

- For an unbounded set A it holds that $\operatorname{diam}(A) = \infty$.
- The maximum value for a non-closed set does not always exist, but intuitively we can instead consider the diameter of the closure $\overline{A}$. The set A if bounded, if $\operatorname{diam}(\overline{A}) < \infty$.

Existence of extrema

- Let $D \subset \mathbb{R}^n$ be a <u>closed</u> and <u>bounded</u> set and $f : D \to \mathbb{R}$ a <u>continuous</u> function. Then f has both a minimum and a maximum in D. In other words, there exist such points $\mathbf{a} \in D$ and $\mathbf{b} \in D$ that $f(\mathbf{a}) = \min_D f$ and $f(\mathbf{b}) = \max_D f$.
- Note. All three underlined conditions are necessary. For example, there is no maximum of f in D, when
  - D is open, but the maximum would come on the boundary.
  - $D = \mathbb{R}^2$ and $f(x, y) = x^2 + y^2$. Then f increases to infinity on the x-axis as $x \to \infty$, but a maximum value is always finite.
  - f is piecewise defined, with one part defined by a continuous expression outside the point $\mathbf{c}$ (which would be its maximum), and the other part is defined at $\mathbf{c}$ by some smaller value of f.

A necessary condition for local extrema

- Assume the function $f : D \to \mathbb{R}$, where $D \subset \mathbb{R}^n$, has a local extremum $\mathbf{a}$. If all partial derivatives of f exist at $\mathbf{a}$, then they all vanish (=are zero) at $\mathbf{a}$, i.e.

$$\frac{\partial f}{\partial x_i}(\mathbf{a}) = 0, \quad \forall i = 1, 2, \ldots, n.$$

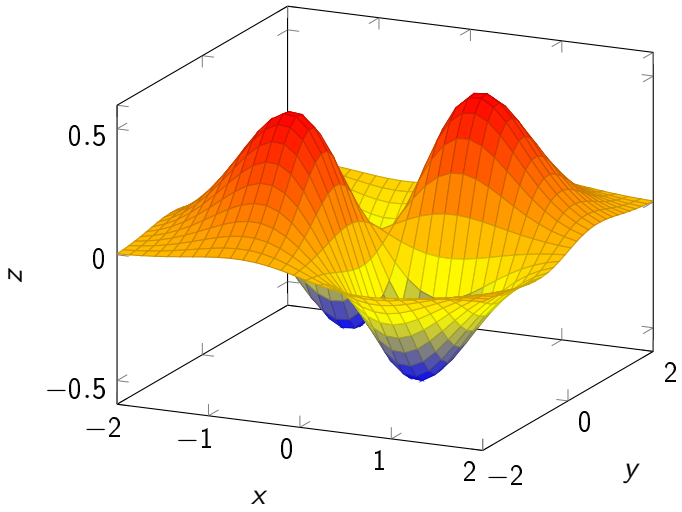
- Global extrema are always also local extrema.
- The above condition does not rule out the existence of extrema in points where one or more partial derivatives do not exist.
- If the domain of f is restricted, i.e. contains fewer values than the equation of f would allow, there might additional extrema on the restricted domain's boundary, if it is included in the domain.

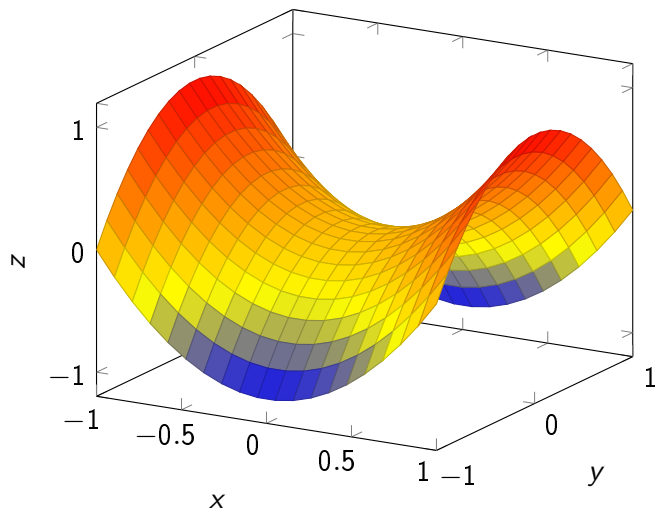
A critical point

- The point **a** is a **critical point** of the function $f : D \to \mathbb{R}$, where $D \subset \mathbb{R}^n$, if all partial derivatives vanish in **a** or at least one of the partial derivatives does not exist in **a**,
- f all partial derivatives of $f : \mathbb{R}^2 \to \mathbb{R}$ vanish in a critical point (a, b), geometrically it means that the tangent plane that is tangent to the surface $z = f(x, y)$ at $(a, b, f(a, b))$, is horizontal, i.e. of the form $z = c$, where $c \in \mathbb{R}$ is a constant.
- Even though all partial derivatives would vanish in a point, it does not mean the point would be an extremum.
  Non-extremum points where all partials vanish are called **saddle points**.

A saddle point

- The graph of $f(x,y) = xye^{1-x^2-y^2}$ on the left has the two local maxima, two local minima and a saddle point at the origin (located between the two "hills")
- On the right is the graph of $f(x,y) = x^2 - y^2$, whose only critical point $(0,0)$ is a saddle point. The surface looks a bit like a saddle.





Procedure for finding global extrema

Let $D \subset \mathbb{R}^n$ be a closed and bounded set and $f : D \to \mathbb{R}$ a continuous function. Then f has both a minimum and a maximum in D. If $f(\mathbf{a})$ is a global minimum value or maximum value of f, then $\mathbf{a}$ is

1. an interior point of D, where all partial derivatives of f vanish,
2. an interior point of D, where f does not have all partial derivatives, or
3. a boundary point of D.

The procedure in other words

- Find those critical points for which all partial derivatives vanish. Check that the found points are within D.
- Find those critical points for which a partial derivative does not exist. For example, a division by zero in the partial derivative could cause this. Check that the found points are within D.
- Boundary: Substitute one boundary curve at a time into the function (like when showing a limit does not exist).

  $\longrightarrow$ You'll get a single variable function, whose extrema you can find in a similar way you did on a previous course, i.e. from the zeros of the derivative, the points where the derivative does not exist or the end points of the domain (which is an interval) Check that the found points are on the interval.

Notes on finding extrema

- The corner points of certain boundaries (like triangles, rectangles etc.) are automatically suspects for extrema, since they are the end points to boundary curves.
- You can find some point of interest from the boundaries that are smooth (like circles, ellipses etc.) by substituting, for example, a parametrization of the boundary curve. Then check the end points given by the parameter interval.
- If you're using a parametrization for finding points on the boundary, the parametrization should be simple, which means that no two parameter values should give the same point except if it is the initial point and at the same time the terminal point of the parametric curve.

When does a function not have extrema?

- Let D be any set in what follows.
- If f increases to infinity along any curve in D, then the function f does not have a maximum. Similarly, if f decreases to minus infinity, there's no minimum.
- For example, the function $f(x,y) = x^4 + y^4 - 3x^2y^2$ considered on the x-axis has the form $f(x,0) = x^4$. When $x \to \infty$, then $f(x,0) \to \infty$ and therefore $f(x,y)$ has no maximum in $\mathbb{R}^2$.
- On the line $y = x$, the same function has the form $f(x,x) = x^4 + x^4 - 3x^4 = -x^4$. When $x \to \infty$, then $f(x,x) \to -\infty$ and therefore $f(x,y)$ has no minimum.

Restriction of the set and inspection of its exterior

- Assume we have found a maximum value M for f in a closed and bounded subset $R \subset D$ of the domain D.
- If we can prove that outside of R all function values are smaller than M, then $\max_D f = M$.
- Let us find the maximum of $f(x,y) = x^3 - x^4 + y^3 - y^4$. For all $|x| > 2$ it holds that $x^4 > x^3$, which is the same as $x^3 - x^4 < 0$. Similarly for all $|y| > 2$ we get $y^3 - y^4 < 0$. In fact, we can prove that $f(x,y) < 0$ outside the square

$$A = \left\{(x,y) \in \mathbb{R}^2 : -2 \leq x \leq 2, -2 \leq y \leq 2\right\}.$$

- The maximum value of f in the square A is $f\left(\frac{3}{4}, \frac{3}{4}\right) = \frac{27}{128}$. Because $f(x,y) < 0$ for $(x,y) \notin A$, then $\max_{\mathbb{R}^2} f = \frac{27}{128}$.

Finding the local extrema in a subset of the xy-plane

- Assume the third order partial derivatives of $f(x,y)$ are continuous in a neighbourhood of (a,b). Additionally denote

$$\Delta = f_{xx} f_{yy} - f_{xy}^2.$$

- If $\Delta(a,b) > 0$ and $f_{xx}(a,b) > 0$, then (a,b) is a local minimum.
- If $\Delta(a,b) > 0$ and $f_{xx}(a,b) < 0$, then (a,b) is a local maximum
- If $\Delta(a,b) < 0$, then (a,b) is a saddle point (not an extremum)
- If $\Delta(a,b) = 0$, this test is inconclusive and you should find some other way of telling if (a,b) is an extremum.

Differentiability of single variable functions

- The function $f:(c,d)\to\mathbb{R}$ is differentiable at $a\in\mathbb{R}$, if there exists

$$f'(a)=\lim_{h\to 0}\frac{f(a+h)-f(a)}{h}$$

$$\Leftrightarrow \lim_{h\to 0}\left(\frac{f(a+h)-f(a)}{h}-f'(a)\right)=0.$$

- Denote the expression in parenthesis by $\varepsilon(h)$. Solve $f(a+h)$.

$$\underbrace{f(a+h)}_{\text{exact value}}=\underbrace{f(a)+f'(a)h}_{\text{approximation}}+\underbrace{h\varepsilon(h)}_{\text{error term}},$$

where $\varepsilon(h)\to 0$ as $h\to 0$.

Differentiability of single variable functions

- The same idea can be reversed. When the latter equation holds, the function is differentiable. Therefore the following theorem is valid.
- The function $f:(c,d)\to\mathbb{R}$ is differentiable at $a\in(c,d)$, if and only if there exists $A\in\mathbb{R}$ and a function $\varepsilon:\mathbb{R}\to\mathbb{R}$ such that $\lim\limits_{h\to 0}\varepsilon(h)=0$ and

$$f(a+h)=f(a)+Ah+h\varepsilon(h)$$

for all $h\in\mathbb{R}$ (whose absolute value is small). Then $A=f'(a)$.

Linearization of single variable functions

- Omit the error term $h\varepsilon(h)$ from the previous equation and denote $x = a + h$, i.e. $h = x - a$. We get the approximation

$$f(x) \approx f(a) + f'(a)(x - a).$$

- The graph of the function on the left hand side,

$$T(x) = f(a) + f'(a)(x - a),$$

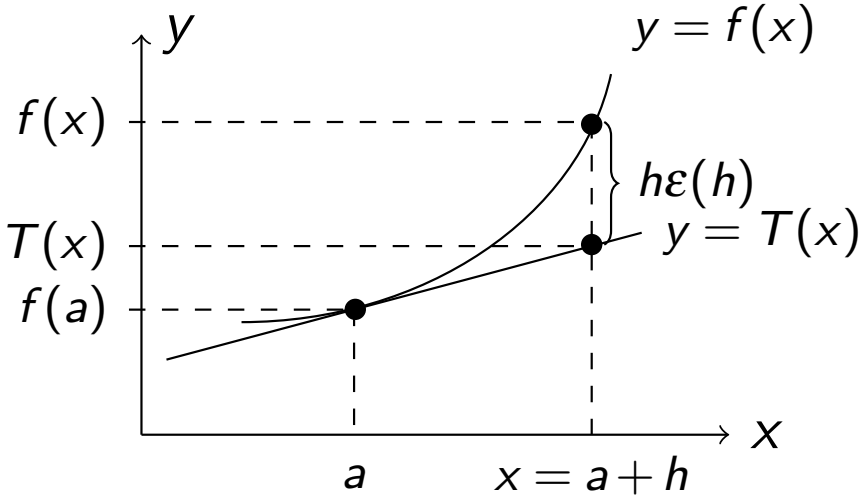
  is the tangent line of f at $(a, f(a))$.
- Then $f(x) \approx T(x)$ is the **linear approximation** or **tangent line approximation** of f and $T(x)$ is the **linearization** of f.

Error in the linearization of single variable functions

- Approximating $f(x) \approx T(x)$, we make the error $f(x) - T(x) = h\varepsilon(h)$. It holds for the error that

$$\frac{\text{error}}{\text{change of } x} = \frac{h\varepsilon(h)}{h} = \varepsilon(h) \to 0, \quad \text{when } h \to 0.$$

- An error made near the approximation point a is therefore negligible relative to the distance from a.



Comparing single variable and multivariable differentiability

- For a single variable functions, differentiability is more or less equivalent to the fact that the derivative exists. It always implies continuity.
- For a two variable function, continuity is not certain even if both partial derivatives exist, because they tell about the behaviour of the function only in the directions of the coordinate axes.
- The definition of differentiability of multivariable functions given on the next page implies continuity. Note that multivariable differentiability is a stronger condition than just the partial derivatives existing, and it gives us certainty of the function behaving "well" in some sense.

Differentiability of two variable functions

- The function $f : D \to \mathbb{R}$, $D \subset \mathbb{R}^2$, is **differentiable at** $(a, b) \in D$, if both partial derivatives of f exist and if

$$f(a+h, b+k) = f(a,b) + \frac{\partial f}{\partial x}(a,b)\,h + \frac{\partial f}{\partial y}(a,b)\,k + \varepsilon(h,k)\,\|(h,k)\|,$$

  where $\varepsilon(h,k) \to 0$, when $(h,k) \to (0,0)$.

- Further, the function f is **differentiable**, if it is differentiable at every point of the domain.

Differentiability of two variable functions

- If we use the notations $a+h=x$ and $b+k=y$, and

$$T(x,y)=f(a,b)+\frac{\partial f}{\partial x}(a,b)(x-a)+\frac{\partial f}{\partial y}(a,b)(y-b),$$

then $f(x,y)\approx T(x,y)$ is the **linear approximation** or **tangent plane approximation** of f and $T(x,y)$ the **linearization** of f.

- For example, the linearization of $f(x,y)=\dfrac{x-y}{x+y}$ in the neighbourhood of $(1,2)$ is

$$T(x,y)=-\frac{1}{3}+\frac{4}{9}(x-1)-\frac{2}{9}(y-2).$$

Error in the linearization of two variable functions

- Approximating $f(x,y) \approx T(x,y)$, we make the error $\varepsilon(h,k)\|(h,k)\|$. Its ratio with the distance from (a,b) is

$$\frac{\varepsilon(h,k)\|(h,k)\|}{\|(h,k)\|} = \varepsilon(h,k) \to 0, \quad \text{when } (h,k) \to (0,0).$$

- Proving differentiability we can done by checking that

$$\begin{aligned}\varepsilon(h,k) &= \frac{f(a+h,b+k) - T(a,b)}{\|(h,k)\|} \\ &= \frac{f(a+h,b+k) - f(a,b) - \frac{\partial f}{\partial x}(a,b)\,h - \frac{\partial f}{\partial y}(a,b)\,k}{\sqrt{h^2+k^2}}\end{aligned}$$

is zero, when $(h,k) \to (0,0)$ (but there's an easier way).

Derivative matrix

- Let $f : D \to \mathbb{R}$, where $D \subset \mathbb{R}^n$, have all partial derivatives at $\mathbf{x}$. Then the **derivative matrix** of f at the point $\mathbf{x}$ is the row vector of all partial derivatives

$$f'(\mathbf{x}) = \begin{bmatrix} \dfrac{\partial f}{\partial x_1}(\mathbf{x}) & \dfrac{\partial f}{\partial x_2}(\mathbf{x}) & \cdots & \dfrac{\partial f}{\partial x_n}(\mathbf{x}) \end{bmatrix}.$$

- For example, the derivative matrix of the function $f(x,y,z) = 2x^2 + 3xy + 4x^2z + 2yz^2$ is

$$f'(x,y,z) = \begin{bmatrix} 4x + 3y + 8xz & 3x + 3z^2 & 4x^2 + 4yz \end{bmatrix}.$$

Derivative matrix in the definition of differentiability

- Denote $\mathbf{x} = (x, y)$, $\mathbf{a} = (a, b)$ and $\mathbf{h} = (h, k)$. If $\mathbf{x} = \mathbf{a} + \mathbf{h}$, then the equation in the definition of differentiability is

$$f(\mathbf{a} + \mathbf{h}) = f(\mathbf{a}) + f'(\mathbf{a})\mathbf{h} + \varepsilon(\mathbf{h})\|\mathbf{h}\|,$$

  where $\mathbf{h}$ is a column vector.

- If $\mathbf{h} = \begin{bmatrix} h_1 & h_2 & \cdots & h_n \end{bmatrix}^T$ is an n-vector, then

$$f'(\mathbf{a})\mathbf{h} = \begin{bmatrix} \frac{\partial f}{\partial x_1}(\mathbf{a}) & \frac{\partial f}{\partial x_2}(\mathbf{a}) & \cdots & \frac{\partial f}{\partial x_n}(\mathbf{a}) \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \\ \vdots \\ h_n \end{bmatrix}$$

$$= \frac{\partial f}{\partial x_1}(\mathbf{a})\, h_1 + \frac{\partial f}{\partial x_2}(\mathbf{a})\, h_2 + \cdots + \frac{\partial f}{\partial x_n}(\mathbf{a})\, h_n.$$

Differentiability of multivariable functions

- The function $f : D \to \mathbb{R}$, where $D \subset \mathbb{R}^n$, is **differentiable at** $\mathbf{a} \in \mathbb{R}^n$, if all n partial derivatives of f exist at $\mathbf{a}$ and if

$$f(\mathbf{a}+\mathbf{h}) = f(\mathbf{a}) + f'(\mathbf{a})\mathbf{h} + \varepsilon(\mathbf{h})\|\mathbf{h}\|,$$

  where $\varepsilon(\mathbf{h}) \to 0$, when $\mathbf{h} \to \mathbf{0}$. Further, f is **differentiable**, if it is differentiable at every point of the domain D.

- The **linearization** of f at the point $\mathbf{a}$ is $T : D \to \mathbb{R}$,

$$T(\mathbf{x}) = f(\mathbf{a}) + f'(\mathbf{a})(\mathbf{x}-\mathbf{a})$$

  and $f(\mathbf{x}) \approx T(\mathbf{x})$ is the **linear approximation** of f at $\mathbf{a}$.

Proving differentiability

- If the function $f : D \to \mathbb{R}$, where $D \subset \mathbb{R}^n$, is differentiable at $\mathbf{a} \in \mathbb{R}^n$, then f is contiuous at $\mathbf{a}$.
- Therefore it seems to be important that we are able to prove the differentiability. However, using the definition to prove it might not be easy.
- If $f : D \to \mathbb{R}$, where $D \subset \mathbb{R}^n$, has continuous partial derivatives, then f is **continuously differentiable**.
- A continuously differentiable function is differentiable.
- Checking whether the partial derivatives are continuous or not is a much easier way to prove differentiability than calculating the limit of $\varepsilon(\mathbf{h})$.

Linearization of a three variable function

- For a three variable function f, denote $\mathbf{x} = (x, y, z)$, $\mathbf{a} = \mathbf{x}_0 = (x_0, y_0, z_0)$ and $\mathbf{h} = \mathbf{x} - \mathbf{x}_0 = (\triangle x, \triangle y, \triangle z)$. Then

$$\begin{cases} x = x_0 + \triangle x, \\ y = y_0 + \triangle y, \\ z = z_0 + \triangle z. \end{cases}$$

  These can be interpreted so that x_0, y_0 and z_0 are measured values, and $\triangle x$, $\triangle y$ and $\triangle z$ are measurement errors.

- The linearization is then

$$T(x, y, z) = f(x_0, y_0, z_0) + \frac{\partial f}{\partial x}(x_0, y_0, z_0)\triangle x + \frac{\partial f}{\partial y}(x_0, y_0, z_0)\triangle y + \frac{\partial f}{\partial z}(x_0, y_0, z_0)\triangle z.$$

Effect of measurement errors

- Let's interpret the previous page's notations as measured values and measurement errors. Denote additionally the difference of the exact function value and the value calculated with the measured inputs

$$\triangle f = f(x, y, z) - f(x_0, y_0, z_0).$$

- If the measurement errors $\triangle x$, $\triangle y$ and $\triangle z$ are small, then the value of the function f is close to the linear approximation, i.e. $f(x, y, z) \approx T(x, y, z)$. Then we can approximate

$$\triangle f \approx \frac{\partial f}{\partial x}(x_0, y_0, z_0)\triangle x + \frac{\partial f}{\partial y}(x_0, y_0, z_0)\triangle y + \frac{\partial f}{\partial z}(x_0, y_0, z_0)\triangle z.$$

Estimation of maximum error

- If we know that in a certain measurement, the maximum errors have the upper bounds $|\triangle x| \leq D_1$, $|\triangle y| \leq D_2$ and $|\triangle z| \leq D_3$, then the triangle inequality gives us an approximation for the maximal error of the function f,

$$
\begin{aligned}
|\triangle f| &\lesssim \left|\frac{\partial f}{\partial x}(x_0, y_0, z_0)\right| |\triangle x| + \left|\frac{\partial f}{\partial y}(x_0, y_0, z_0)\right| |\triangle y| \\
&\quad + \left|\frac{\partial f}{\partial z}(x_0, y_0, z_0)\right| |\triangle z| . \\
&\leq \left|\frac{\partial f}{\partial x}(x_0, y_0, z_0)\right| D_1 + \left|\frac{\partial f}{\partial y}(x_0, y_0, z_0)\right| D_2 \\
&\quad + \left|\frac{\partial f}{\partial z}(x_0, y_0, z_0)\right| D_3 .
\end{aligned}
$$